# • Algoritmi Valori și Vectori Proprii

## *Localizarea cercurilor lui Gershgorin

$\lambda(A) \in$ mulțimea discurilor $D_i = \{z \in \mathbb{C} \mid |z - a_{ii}| \leq \sum_{\substack{j=1 \\ j \neq i}}^{n} |a_{ij}|\}$

$a_{ii}$ ↑ elem pe diag

$\sum_{j \neq i}$ suma elem. de pe linie.

- fiecare $\lambda \in D_i$, dar pot exista $D_i$ fără valori $\lambda$
- se știe dinainte are o singură valoare $\lambda$
- o mulțime de K discuri intersectate conține K valori $\lambda$

Examen 2021 (Subiectul 2)

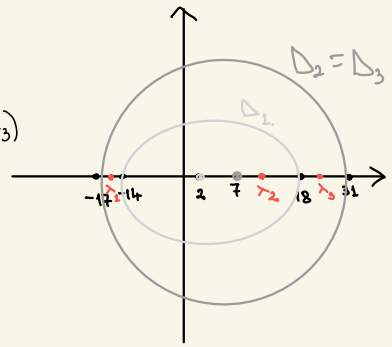
$A = \begin{bmatrix} 2 & 8 & 8 \\ 8 & 7 & -16 \\ 8 & -16 & 7 \end{bmatrix}$

$\det(A - \lambda I_3) = \begin{vmatrix} 2-\lambda & 8 & 8 \\ 8 & 7-\lambda & -16 \\ 8 & -16 & 7-\lambda \end{vmatrix} \Rightarrow \lambda \in \{\underbrace{\frac{-\sqrt{1633}-7}{2}}_{\lambda_1}; \underbrace{\frac{\sqrt{1633}-7}{2}}_{\lambda_2}; \underbrace{23}_{\lambda_3}\}$

$D_1: |z-2| \leq 16 \; (\lambda_2)$

$D_2: |z-7| \leq 24$

$D_3: |z-7| \leq 24$ $\Big\}$ $(\lambda_1, \lambda_3)$



## * M.P.D. (Metoda puterii directe)

**Date input**
- A - o matrice patratică
- $y_0$ - aprox inițială vect. propriu
- max_iter - nr. max pași
- tol - toleranță

**Date output**
- lambda - val. proprie maximă
- y - vect. propriu maxim

**Convergență** [matrice diagonalizabilă și stabilă numeric; trace spectrală e dată de o singură valoare.

**OCTAVE:**

```
function [lambda, y, iter] = MPD(A, y0, max_iter, tol)
    y = y0;
    for i = 1:max_iter
        z = A*y;
        y = z / norm(z);
        lambda = y' * A * y;
        if (abs(A*y - lambda*y) < tol)
            break;
        endif
    endfor
endfunction
```

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### EXAMEN 2020 (SUBIECTUL 1)

Ex. $A = \begin{bmatrix} -1 & 4 \\ 4 & 5 \end{bmatrix}$

a) $\det(A - \lambda I_2) = 0 \Leftrightarrow \begin{vmatrix} -1-\lambda & 4 \\ 4 & 5-\lambda \end{vmatrix} = 0 \Leftrightarrow$

$\Leftrightarrow (\lambda+1)(\lambda-5) - 16 = 0 \Leftrightarrow \lambda^2 - 4\lambda - 5 - 16 = 0 \Leftrightarrow$

$\Leftrightarrow \lambda^2 - 4\lambda - 21 = 0 \Leftrightarrow \lambda \in \{-3; 7\} \Rightarrow \lambda(A) = \{-3, 7\}$

b) MPD pt A cu $x_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$y_0 = x_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$z_1 = A \cdot y_0 = \begin{bmatrix} -1 & 4 \\ 4 & 5 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 9 \end{bmatrix}$

$y_1 = \frac{z_1}{\|z_1\|_2} = \frac{1}{3\sqrt{10}} \cdot \begin{bmatrix} 3 \\ 9 \end{bmatrix} = \frac{1}{\sqrt{10}} \cdot \begin{bmatrix} 1 \\ 3 \end{bmatrix}$

$\lambda_1 = y_1^T \cdot A \cdot y_1 = \frac{1}{\sqrt{10}} \cdot \begin{bmatrix} 1 & 3 \end{bmatrix} \cdot \begin{bmatrix} -1 & 4 \\ 4 & 5 \end{bmatrix} \cdot \frac{1}{\sqrt{10}} \cdot \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \frac{68}{10} = 6{,}8$

$z_2 = A \cdot y_1 = \begin{bmatrix} -1 & 4 \\ 4 & 5 \end{bmatrix} \cdot \frac{1}{\sqrt{10}} \cdot \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \frac{1}{\sqrt{10}} \cdot \begin{bmatrix} 11 \\ 19 \end{bmatrix}$

$y_2 = \frac{z_2}{\|z_2\|_2} = \frac{\sqrt{10}}{482} \cdot \frac{1}{\sqrt{10}} \cdot \begin{bmatrix} 11 \\ 19 \end{bmatrix} = \frac{1}{\sqrt{482}} \cdot \begin{bmatrix} 11 \\ 19 \end{bmatrix}$

$\lambda_2 = y_2^T \cdot A \cdot y_2 = \frac{1}{\sqrt{482}} \cdot \begin{bmatrix} 11 & 19 \end{bmatrix} \cdot \begin{bmatrix} -1 & 4 \\ 4 & 5 \end{bmatrix} \cdot \frac{1}{\sqrt{482}} \cdot \begin{bmatrix} 11 \\ 19 \end{bmatrix} = \frac{3356}{482} = 6{,}96$

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### EXAMEN 2022 (SUBIECTUL 1)

Fie $A = \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix}$

a) MPD pt. A cu $y_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$z_1 = A \cdot y_0 = \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$y_1 = \frac{z_1}{\|z_1\|_2} = \frac{1}{\sqrt{5}} \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$\lambda_1 = y_1^T \cdot A \cdot y_1 = \frac{1}{\sqrt{5}} \cdot \begin{bmatrix} 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix} \cdot \frac{1}{\sqrt{5}} \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \frac{10}{5} = 2$

$z_2 = A \cdot y_1 = \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix} \cdot \frac{1}{\sqrt{5}} \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \frac{1}{\sqrt{5}} \cdot \begin{bmatrix} 0 \\ 5 \end{bmatrix}$

$y_2 = \frac{z_2}{\|z_2\|_2} = \frac{1}{\sqrt{5}} \cdot \frac{1}{\sqrt{5}} \cdot \begin{bmatrix} 0 \\ 5 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$\lambda_2 = y_2^T \cdot A \cdot y_2 = \begin{bmatrix} 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 3$

$z_3 = A \cdot y_2 = \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$

$y_3 = \frac{z_3}{\|z_3\|_2} = \frac{1}{\sqrt{10}} \cdot \begin{bmatrix} -1 \\ 3 \end{bmatrix}$

$\lambda_3 = y_3^T \cdot A \cdot y_3 = \frac{1}{\sqrt{10}} \cdot \begin{bmatrix} -1 & 3 \end{bmatrix} \cdot \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix} \cdot \frac{1}{\sqrt{10}} \cdot \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \frac{35}{10} = 3{,}5$

d) $\det(A - \lambda I_2) = 0 \Leftrightarrow \begin{vmatrix} 2-\lambda & -1 \\ -1 & 3-\lambda \end{vmatrix} = 0 \Leftrightarrow$

$\Leftrightarrow (\lambda-2)(\lambda-3) - 1 = 0 \Leftrightarrow \lambda^2 - 5\lambda + 5 = 0 \Leftrightarrow$

$\Leftrightarrow \lambda^2 - 5\lambda + 5 = 0 \Rightarrow \lambda \in \left\{ \frac{5-\sqrt{5}}{2}; \frac{5+\sqrt{5}}{2} \right\} \Rightarrow \lambda(A) = \left\{ \frac{5-\sqrt{5}}{2}, \frac{5+\sqrt{5}}{2} \right\}$

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## * Calculul celei mai mici valori proprii

**OCTAVE:**

```
function [lambda, y, iter] = MPD(A, y0, max_iter, tol)
    A = inv(A);
    y = y0;
    for i = 1:max_iter
        z = A*y;
        y = z / norm(z);
        lambda = y' * A * y;
        if (abs(A*y - lambda*y) < tol)
            break;
        endif
    endfor
    A = inv(A);
    lambda = 1/lambda;
endfunction
```

## * M.P.I. (Metoda puterii Inverse)

**Date input**
- $A$ - o matrice pătratică
- $y_0$ - aprox inițială vect. propriu
- $\mu$ - un factor.
- max_iter - nr. max pași
- tol - toleranță

**Date output**
- lambda - val. proprie maximă
- $y$ - vect. propriu maxim

**Convergență** [matrice diagonalizabilă și stabilă numeric
trasa spectrală e dată de o singură valoare.

**OCTAVE:**

```
function [lambda, y, iter] = MPI (A, y0, miu, max_iter, tol)
    [n,n] = size(A);
    y = y0;
    for i = 1: max_iter
        C = A - miu * eye(n);
        [C,z] = eliminare_Gauss(C,y);
        z = SST(C,z);
        y = z / norm(z);
        lambda = y' * A * y;
        miu = lambda;
        if (abs(A*y - lambda*y) < tol)
            break;
        endif
    endfor
endfunction
```

## * Descompunerea valorilor singulare (S.V.D.)

$A = U \cdot \Sigma \cdot V^T$

**PAS 1: calcul $\Sigma$**
- calculez $A^T \cdot A$
- calculez $\lambda$ pentru $A^T \cdot A$
- calculez $\sigma = \sqrt{\lambda}$
- pun $\sigma$ descrescător pe diagonala $\Sigma$ (cât se poate)

**PAS 2: calcul V**
- folosim relația $A^T \cdot A \cdot v_i = \lambda \cdot v_i \Rightarrow$ vectorii proprii
- îi pun vectorii proprii v pe coloană (în ordinea descrescătoare a valorilor proprii)

**PAS 3: calcul U**

$u_i = \frac{1}{\sigma} \cdot v_i$ (pt $\min(m,n)$ vectori).

* restul de u se calculează cu G.S.

**OCTAVE**

```
SVD(A)
```

**EXAMEN 2021 (SUBIECTUL 1)**

$A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \end{bmatrix}$

$A^T \cdot A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 1 & 2 \\ 3 & 2 & 5 \end{bmatrix}$

$\det(A^T A - \lambda I_3) = \begin{vmatrix} 2-\lambda & 1 & 3 \\ 1 & 1-\lambda & 2 \\ 3 & 2 & 5-\lambda \end{vmatrix} \Rightarrow \lambda \in \{0, -\sqrt{13}+4, \sqrt{13}+4\}$.

$\Sigma = \begin{pmatrix} \sqrt{\sqrt{13}+4} & 0 & 0 \\ 0 & \sqrt{-\sqrt{13}+4} & 0 \end{pmatrix}$

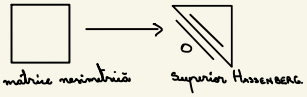
...

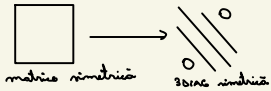
# * Matrici asemenea

A, B - asemenea $\Leftrightarrow \exists$ T - nesingulară a.î. $B = T^{-1} \cdot A \cdot T$.

## • Aducere la Forma Superior Hessenberg

∀ matrice pătratică poate fi adusă la forma superior Hessenberg
(transformare de asemănare ortogonală)
$H_0 = P \cdot A \cdot P^T$, P - matrice ortogonală




Exemplu Rezolvat

$A = \begin{pmatrix} 0 & 3 & 4 \\ 3 & 2 & 3 \\ 4 & 3 & 1 \end{pmatrix}$

$v_1 = \begin{pmatrix} 3 \\ 4 \end{pmatrix} \Rightarrow \overline{v}_1 = \text{semn}(3) \cdot \|v_1\|_2 = 5$

$\tilde{v}_1 = v_1 + \begin{pmatrix} \overline{v}_1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 8 \\ 4 \end{pmatrix}$

$H_1 = I_2 - \frac{2 \cdot \tilde{v}_1 \cdot \tilde{v}_1^t}{\tilde{v}_1^t \cdot \tilde{v}_1} = \begin{pmatrix} -0,6 & -0,8 \\ -0,8 & 0,6 \end{pmatrix}$

$\tilde{H}_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & & \\ 0 & & H_1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -0,6 & -0,8 \\ 0 & -0,8 & 0,6 \end{pmatrix}$

$Q_1 = \tilde{H}_1^T = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -0,6 & -0,8 \\ 0 & -0,8 & 0,6 \end{pmatrix}$

$R_1 = \tilde{H}_1 \cdot A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -0,6 & -0,8 \\ 0 & -0,8 & 0,6 \end{pmatrix} \begin{pmatrix} 0 & 3 & 4 \\ 3 & 2 & 3 \\ 4 & 3 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 3 & 4 \\ -5 & -\frac{18}{5} & -\frac{23}{5} \\ 0 & \frac{1}{5} & \frac{3}{5} \end{pmatrix}$

$A_1 = R_1 \cdot \tilde{H}_1^T = \begin{pmatrix} 0 & -5 & 0 \\ -5 & \frac{106}{25} & \frac{33}{25} \\ 0 & \frac{33}{25} & -\frac{31}{25} \end{pmatrix}$

## • Forma Schur

∀ matrice pătratică poate fi adusă la forma Schur (cvasi superior triunghiulară)
(transformare de asemănare ortogonală)
$S = P^T \cdot A \cdot P$, P - matrice ortogonală

## * ALGORITM QR fără deplasare

OCTAVE

```
function[Hv] = QR_fără_deplasare(A, max_iter).
    Hv = A;
    for K=0:max_iter
        [Q, R] = householder(Hv);
        Hv = R·Q;
    endfor.
endfunction.
```

## * ALGORITM QR cu deplasare - pt matrici nesimetrice

OCTAVE

```
function[Hv] = QR_ cu_deplasare(A, miu, max_iter)
    [n, n] = size(A)
    Hv = A - miu * eye(n)
    for K=0:max_iter
        miu = Hv(n,n);
        [Q, R] = householder(Hv);
        Hv = R·Q + miu * eye(n);
    endfor.
endfunction.
```

## * ALGORITM QR cu deplasare - pt matrici simetrice

OCTAVE

```
function[Hv] = QR_ cu_deplasare(A, miu, max_iter)
    [n, n] = size(A)
    Hv = A - miu * eye(n)
    for K=0:max_iter
        miu = MPD(Hv, 2:norm(n), 0,001, 100);
        [Q, R] = householder(Hv);
        Hv = R·Q + miu * eye(n);
    endfor.
endfunction.
```

# INTERPOLARE

Fie $f:[x_0;x_n]\to\mathbb{R}$, cunoscută doar în punctele $x_0,\dots,x_n$ prin valorile $f(x_0), f(x_1),\dots,f(x_n)$.

$\{x_0,\dots,x_n\}$ - suportul interpolării

! $P_n(x) = a_0u_0(x) + a_1u_1(x) + \dots + a_nu_n(x)$

$P_n$: polinom de interpolare de grad $n$.

$u_0,\dots,u_n$: bază interpolării (se dă în problemă)

$f(x)\simeq P_n(x)$, $x\in[x_0,x_n]$ - {suportul interpolării}.

* pentru a afla $a_0,\dots,a_n$ folosim $f(x)=P_n(x)$, $x\in$ {suportul interpolării} (condiție de interpolare)

## ! POLINOAME BERNSTEIN DE GRADUL N

$B_{i,n}(t) = C_n^i(1-t)^{n-i}t^i$, $i=0:n$, $t$ - parametru $\in[0,1]$.

**n = 1**

$B_{0,1}(t) = C_1^0(1-t)^1\cdot t^0 = 1-t$.

$B_{1,1}(t) = C_1^1(1-t)^0\cdot t^1 = t$

**n = 2**

$B_{0,2}(t) = C_2^0(1-t)^2\cdot t^0 = (1-t)^2$

$B_{1,2}(t) = C_2^1(1-t)^1\cdot t^1 = 2(1-t)\cdot t$.

$B_{2,2}(t) = C_2^2(1-t)^0\cdot t^2 = t^2$

**n = 3**

$B_{0,3}(t) = C_3^0(1-t)^3t^0 = (1-t)^3$

$B_{1,3}(t) = C_3^1(1-t)^2\cdot t^1 = 3(1-t)^2\cdot t$.

$B_{2,3}(t) = C_3^2(1-t)^1\cdot t^2 = 3(1-t)\cdot t^2$

$B_{3,3}(t) = C_3^3(1-t)^0\cdot t^3 = t^3$.

## * Interpolare Polinomială (Vandermonde)

- Baza de interpolare: $\{1, X, X^2, \dots, X^n\}$
- $P_n(X) = a_0 + a_1 X + a_2 X^2 + \dots + a_n X^n$
- Condiție interpolare: $f(X) = P_n(X)$, $X \in \{\text{suportul interpolării}\}$.

$$\begin{cases} a_0 + a_1 x_0 + a_2 x_0^2 + \dots + a_n x_0^n = f(x_0) \\ a_0 + a_1 x_1 + a_2 x_1^2 + \dots + a_n x_1^n = f(x_1) \\ \vdots \\ a_0 + a_1 x_n + a_2 x_n^2 + \dots + a_n x_n^n = f(x_n) \end{cases} \Leftrightarrow$$

$$\Leftrightarrow \begin{bmatrix} 1 & x_0 & x_0^2 & \dots & x_0^n \\ 1 & x_1 & x_1^2 & \dots & x_1^n \\ \vdots & & & & \\ 1 & x_n & x_n^2 & \dots & x_n^n \end{bmatrix} \cdot \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} f(x_0) \\ f(x_1) \\ \vdots \\ f(x_n) \end{bmatrix}$$

matrice VANDERMONDE (nu neapărat condiționată).

### Exemplu Rezolvat.

| $x$ | 1 | 2 | 3 |
|---|---|---|---|
| $f(x)$ | 1 | 8 | 27 |

$P_2(X) = a_0 + a_1 X + a_2 X^2$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 8 \\ 27 \end{bmatrix} \Leftrightarrow \begin{cases} a_0 + a_1 + a_2 = 1 \\ a_0 + 2a_1 + 4a_2 = 8 \\ a_0 + 3a_1 + 9a_2 = 27 \end{cases} \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} a_0 = 6 \\ a_1 = -11 \\ a_2 = 6 \end{cases} \Rightarrow P_2(X) = 6 - 11X + 6X^2$$

# Interpolare Lagrange

- Baza de interpolare: $\{L_0(x), L_1(x), \dots, L_n(x)\}$ (multiplicatorii Lagrange)

$$L_0(x) = \frac{(x-x_1)(x-x_2)\dots(x-x_n)}{(x_0-x_1)(x_0-x_2)\dots(x_0-x_n)}$$

$$L_1(x) = \frac{(x-x_0)(x-x_2)\dots(x-x_n)}{(x_1-x_0)(x_1-x_2)\dots(x_1-x_n)}$$

$$\vdots$$

$$L_k(x) = \prod_{\substack{i=0 \\ i \neq k}}^{n} \frac{x-x_i}{x_k-x_i}$$

- $P_n(x) = \sum_{k=0}^{n} \underbrace{f(x_k)}_{a_k} \cdot L_k(x) = \underbrace{f(x_0)}_{a_0} L_0(x) + \underbrace{f(x_1)}_{a_1} L_1(x) + \dots + \underbrace{f(x_n)}_{a_n} L_n(x)$.

Exemplu: funcția Runge (rezultate nesatisfăcătoare)

$f: [-1,1] \to \mathbb{R}$, $f(x) = \frac{1}{1+25x^2}$

## Exemplu Rezolvat

| $x$ | 1 | 2 | 3 |
|---|---|---|---|
| $f(x)$ | 1 | 8 | 27 |

$$P_2(x) = a_0 L_0(x) + a_1 L_1(x) + a_2 L_2(x)$$

$$L_0(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} = \frac{(x-2)(x-3)}{(1-2)(1-3)} = \frac{(x-2)(x-3)}{2}$$

$$L_1(x) = \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} = \frac{(x-1)(x-3)}{(2-1)(2-3)} = -(x-1)(x-3)$$

$$L_2(x) = \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} = \frac{(x-1)(x-2)}{(3-1)(3-2)} = \frac{(x-1)(x-2)}{2}$$

$$P_2(x) = \frac{(x-2)(x-3)}{2} - 8(x-1)(x-3) + \frac{27(x-1)(x-2)}{2}$$

## * Interpolare Newton

- Baza de interpolare: $\{\underbrace{1}_{u_0}, \underbrace{(x-x_0)}_{u_1}, \underbrace{(x-x_0)(x-x_1)}_{u_2}, \dots, \underbrace{(x-x_0)\dots(x-x_{n-1})}_{u_n}\}$
- $P_n(x) = \underbrace{F_0[x_0]}_{a_0} + \underbrace{F_1[x_0,x_1]}_{a_1}\cdot(x-x_0) + \dots + \underbrace{F_n[x_0,\dots,x_n]}_{a_n}\cdot(x-x_0)\dots(x-x_{n-1})$

Diferențe divizate

Grad 0

$F_0[x_0] = f(x_0)$

$F_0[x_1] = f(x_1)$

$F_0[x_2] = f(x_2)$

$\vdots$

$F_0[x_n] = f(x_n)$

Grad 1

$F_1[x_0,x_1] = \dfrac{F_0[x_0]-F_0[x_1]}{x_0-x_1}$

$F_2[x_1,x_2] = \dfrac{F_0[x_1]-F_0[x_2]}{x_1-x_2}$

Grad 2

$F_2[x_0,x_1,x_2]\dots$

## Exemplu Rezolvat (Examen 2020, Subiectul 2)

| $x$ | 4 | -1 | 0 | 18 | 19. |
|---|---|---|---|---|---|
| $f(x)$ | 6 | -3 | 2 | 8 | 5 |

$$P_4(x) = F_0[x_0] + F_1[x_0,x_1]\cdot(x-x_0) + F_2[x_0,x_1,x_2]\cdot(x-x_0)(x-x_1) +$$
$$+ F_3[x_0,x_1,x_2,x_3]\cdot(x-x_0)(x-x_1)(x-x_2) + F_4[x_0,x_1,x_2,x_3,x_4]\cdot(x-x_0)(x-x_1)(x-x_2)(x-x_3)$$

$F_0[x_0]=6$, $F_0[x_1]=-3$, $F_0[x_2]=2$, $F_0[x_3]=8$, $F_0[x_4]=5$

$F_1[x_0,x_1] = \frac{9}{5}$, $F_1[x_1,x_2] = 5$, $F_1[x_2,x_3] = -\frac{1}{3}$, $F_1[x_3,x_4] = -3$

$F_2[x_0,x_1,x_2] = -\frac{4}{5}$, $F_2[x_1,x_2,x_3] = -\frac{14}{57}$, $F_2[x_2,x_3,x_4] = \frac{10}{57}$

$F_3[x_0,x_1,x_2,x_3] = \frac{79}{1995}$, $F_3[x_1,x_2,x_3,x_4] = \frac{6}{285}$

$F_4[x_0,x_1,x_2,x_3,x_4] = \frac{-37}{29925}$

$\dots$

## INTERPOLARE LINIARĂ (SPLINE LINIAR)

$S: [x_0, x_n] \to \mathbb{R}$, $S_i: [x_i, x_{i+1}] \to \mathbb{R}$ $\forall i = 0 : n-1$

$$S(x) = \begin{cases} S_0(x), & x \in [x_0, x_1) \\ S_1(x), & x \in [x_1, x_2] \\ \vdots \\ S_{n-1}(x), & x \in [x_{n-1}, x_n] \end{cases} ; \quad S_i(x) = a_i x + b_i, \ \forall i = 0 : n-1.$$

DETERMINARE $a_i$ și $b_i$ (CONDIȚII INTERPOLARE + RACORDARE)

$$\begin{cases} a_i = \dfrac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i} \\ b_i = \dfrac{x_{i+1} \cdot f(x_i) - x_i \cdot f(x_{i+1})}{x_{i+1} - x_i} \end{cases}, \ \forall i = 0 : n-1.$$

→ INTERPOLARE SPLINE $C_1$ (CUBICĂ DE TIP $C_1$).

$$\begin{cases} S_i(x) = a_i + b_i(x - x_i) + c_i(x - x_i)^2 + d_i(x - x_i)^3 \\ S_i'(x) = b_i + 2c_i(x - x_i) + 3d_i(x - x_i)^2 \end{cases}, \ \forall i = 0 : n-1.$$

→ INTERPOLARE SPLINE $C_2$ (CUBICĂ DE TIP $C_2$)

$$\begin{cases} S_i(x) = a_i + b_i(x - x_i) + c_i(x - x_i)^2 + d_i(x - x_i)^3 \\ S_i'(x) = b_i + 2c_i(x - x_i) + 3d_i(x - x_i)^2 \\ S_i''(x) = 2c_i + 6d_i(x - x_i) \end{cases}, \ \forall i = 0 : n-1.$$

## EXEMPLU REZOLVAT (EXAMEN 2022, SUBIECTUL 2)

| $x$ | 1 | 2 | 5 | 6 |
|---|---|---|---|---|
| $f(x)$ | 1 | -3 | 2 | -2 |

$$S(x) = \begin{cases} S_0(x), & x \in [1;2] \\ S_1(x), & x \in [2;5] \\ S_2(x), & x \in [5;6]. \end{cases} : S_i(x) = a_i x + b_i$$

$S_0(x) = a_0 x + b_0$

$a_0 = \dfrac{f(x_1) - f(x_0)}{x_1 - x_0} = -4.$

$b_0 = \dfrac{x_1 \cdot f(x_0) - x_0 \cdot f(x_1)}{x_1 - x_0} = 5$

$\Rightarrow S_0(x) = -4x + 5$

$S_1(x) = a_1 x + b_1$

$a_1 = \dfrac{f(x_2) - f(x_1)}{x_2 - x_1} = \dfrac{5}{3}$

$b_1 = \dfrac{x_2 \cdot f(x_1) - x_1 \cdot f(x_2)}{x_2 - x_1} = \dfrac{-19}{3}$

$\Rightarrow S_1(x) = \dfrac{5x - 19}{3}.$

$S_2(x) = a_2 x + b_2$

$a_2 = \dfrac{f(x_3) - f(x_2)}{x_3 - x_2} = -4$

$b_2 = \dfrac{x_3 \cdot f(x_2) - x_2 \cdot f(x_3)}{x_3 - x_2} = 22$

$\Rightarrow S(x) = -4x + 22$

* INTERPOLARE SPLINE $C_2$ (CUBICĂ DE TIP $C_2$)

- $S_i(x) = a_i + b_i(x-x_i) + c_i(x-x_i)^2 + d_i(x-x_i)^3$, $\forall i = 0 : n-1$.

DETERMINARE $a_i, b_i, c_i, d_i$.

CONDIȚII DE INTERPOLARE HERMITE.

$S_i(x_i) = f(x_i)$, $\forall i = 0 : n-1$ | $\Rightarrow S_i'(x_i) = f'(x_i)$, $\forall i = 0 : n-1$

$S_{n-1}(x_n) = f(x_n)$ | $S_{n-1}'(x_n) = f'(x_n)$

CONDIȚII DE RACORDARE:

$S_i(x_{i+1}) = S_{i+1}(x_{i+1})$, $\forall i = 0 : n-2$ | $\Rightarrow S_i'(x_{i+1}) = S_{i+1}'(x_{i+1})$, $\forall i = 0 : n-2$

- $S_i(t) = a_i + b_i \ell_i t + c_i \ell_i^2 t^2 + d_i \ell_i^3 t^3$, $\forall i = 0 : n-1$.

$\ell_i$ - lungimea. $\ell_i = [x_i, x_{i+1}] = x_{i+1} - x_i$, $\forall i = 0 : n-1$.

$t$ - parametru $= \frac{x - x_i}{\ell_i}$, $\forall i = 0 : n-1$.

.....

← folosind polinomul Bernstein de gradul 3:

$S_i(t) = a_i'(1-t)^3 + b_i' 3(1-t)^2 t + c_i' 3(1-t) t^2 + d_i' t^3$, $\forall i = 0 : n-1$.

$$\begin{cases} a_i' = f(x_i) \\ b_i' = f(x_i) + \frac{\ell_i}{3} f'(x_i) \\ c_i' = f(x_{i+1}) - \frac{\ell_i}{3} f'(x_{i+1}) \\ d_i' = f(x_{i+1}) \end{cases} \quad \forall i = 0 : n-1.$$

Rămâne de schimbat variabila $t \to x$.

EXEMPLU REZOLVAT.

| $x$ | 1 | 2 | 4 |
|---|---|---|---|
| $f(x)$ | 3 | 4 | 6 |
| $f'(x)$ | 0 | 2 | 5 |

INTERPOLARE:

$S_0(x_0) = f(x_0) = 3$.

$S_0'(x_0) = f'(x_0) = 0$

RACORDARE:

$S_0(x_1) = S_1(x_1) = 4$

$S_0'(x_1) = S_1'(x_1) = 2$

$S_1(x_1) = f(x_1) = 4$.

$S_1'(x_1) = f'(x_1) = 2$.

$S_1(x_2) = S_2(x_2) = 6$.

$S_1'(x_2) = S_2'(x_2) = 5$

$S_2(x_2) = f(x_2) = 6$

$S_2'(x_2) = f'(x_2) = 5$

$S_0(x) = a_0 + b_0(x-x_0) + c_0(x-x_0)^2 + d_0(x-x_0)^3$.

$S_0'(x) = b_0 + 2c_0(x-x_0) + 3d_0(x-x_0)^2$

$S_1(x) = a_1 + b_1(x-x_1) + c_1(x-x_1)^2 + d_1(x-x_1)^3$

$S_1'(x) = b_1 + 2c_1(x-x_1) + 3d_1(x-x_1)^2$.

$S_2(x) = a_2 + b_2(x-x_2) + c_2(x-x_2)^2 + d_2(x-x_2)^3$

$S_2'(x) = b_2 + 2c_2(x-x_2) + 3d_2(x-x_2)^2$.

## Interpolare Spline $C_2$ (Cubică de tip $C_2$).

$S_i(x) = a_i + b_i(x-x_i) + c_i(x-x_i)^2 + d_i(x-x_i)^3, \forall i = 0:n-1.$

Determinare $a_i, b_i, c_i, d_i$.

Condiții de interpolare:

$S_i(x_i) = f(x_i), \forall i = 0:n-1.$

$S_{n-1}(x_n) = f(x_n).$

Condiții de racordare:

$S_i(x_{i+1}) = S_{i+1}(x_{i+1}), \forall i = 0:n-2. \quad (1) \Rightarrow$

$\Rightarrow S_i'(x_{i+1}) = S_{i+1}'(x_{i+1}), \forall i = 0:n-2. \quad (2) \Rightarrow$

$\Rightarrow S_i''(x_{i+1}) = S_{i+1}''(x_{i+1}), \forall i = 0:n-2.$

Spline $C_2$ (cubică de tip $C_2$) natural: $S_0''(x_0) = S_{n-1}''(x_n) = 0$.

Spline $C_2$ (cubică de tip $C_2$) tensionat: $S_0'(x_0) = f'(x_0)$.

$S_{n-1}'(x_n) = f'(x_n)$.

## Exemplu Rezolvat (Examen 2019, Subiectul 5)

| $x$ | 1 | 2 | 3 |
|---|---|---|---|
| $f(x)$ | 5 | -1 | 4 |

Interpolare:

$S_0(x_0) = f(x_0) = 5$

$S_1(x_1) = f(x_1) = -1.$

$S_1(x_2) = f(x_2) = 4.$

Racordare:

$S_0(x_1) = S_1(x_1) = -1.$

$S_0'(x_1) = S_1'(x_1) = 0$

$S_0''(x_1) = S_1''(x_1) = 0$

Natural:

$S_0''(x_0) = S_1''(x_2) = 0$

$S_0(x) = a_0 + b_0(x-x_0) + c_0(x-x_0)^2 + d_0(x-x_0)^3.$

$S_0'(x) = b_0 + 2c_0(x-x_0) + 3d_0(x-x_0)^2$

$S_0''(x) = 2c_0 + 6d_0(x-x_0)$

$S_1(x) = a_1 + b_1(x-x_1) + c_1(x-x_1)^2 + d_1(x-x_1)^3$

$S_1'(x) = b_1 + 2c_1(x-x_1) + 3d_1(x-x_1)^2.$

$S_1''(x) = 2c_1 + 6d_1(x-x_1)$

$S_2(x) = a_2 + b_2(x-x_2) + c_2(x-x_2)^2 + d_2(x-x_2)^3$

$S_2'(x) = b_2 + 2c_2(x-x_2) + 3d_2(x-x_2)^2.$

$S_2''(x) = 2c_2 + 6d_2(x-x_2)$

## Curbe Bezier

Fie $P_0(x_0, y_0), \dots, P_n(x_n, y_n)$.

$P_0, P_n$ - puncte de interpolare.

$P_1, \dots, P_{n-1}$ - puncte de control

Curba Bezier de ordin $N$: $B_n(t) = \sum_{i=0}^{n} P_i \cdot B_{i,n}(t)$.

- $n=1$ **Curba Bezier liniară (2 puncte de intrare)**

$B_1(t) = P_0 \cdot B_{0,1}(t) + P_1 B_{1,1}(t) =$

$= P_0(1-t) + P_1 t, \forall t \in [0,1]$.

- $n=2$ **Curba Bezier pătratică (3 puncte de intrare)**

$B_2(t) = P_0 \cdot B_{0,2}(t) + P_1 \cdot B_{1,2}(t) + P_2 B_{2,2}(t) =$

$= P_0 \cdot (1-t)^2 + P_1 \cdot 2t(1-t) + P_2 \cdot t^2, \forall t \in [0,1]$.

- $n=3$ **Curba Bezier cubic (4 puncte de intrare).**

$B_3(t) = P_0 \cdot B_{0,3}(t) + P_1 \cdot B_{1,3}(t) + P_2 \cdot B_{2,3}(t) + P_3 B_{3,3}(t) =$

$= P_0 \cdot (1-t)^3 + P_1 \cdot 3(1-t)^2 t + P_2 \cdot 3(1-t)t^2 + P_3 \cdot t^3, \forall t \in [0,1]$.

## Exemplu Rezolvat (Examen 2019, Subiectul 6)

$P_0(0,1), P_1(2,4), P_2(1,0)$ - punctele curbei

* punctul aflat pe curba Bezier pt. $t = 0,5$

$B_2(t) = P_0 \cdot (1-t)^2 + P_1 \cdot 2t(1-t) + P_2 \cdot t^2, t \in [0;1]$

$t = 0,5 \Rightarrow B_2(0,5) = (0;1) \cdot \frac{1}{4} + (2;4) \cdot \frac{1}{2} + (1;0) \cdot \frac{1}{4} \Leftrightarrow$

$\Leftrightarrow B_2(0,5) = (0;\frac{1}{4}) + (1;2) + (\frac{1}{4};0) = (\frac{5}{4};\frac{9}{4})$

## * Algoritmul Lui Casteljau

- generarea a unui punct pe o curbă Bezier

Octave

```
function [P_max] = casteljau(nr_puncte, P)
                   (vector 1x2)        (vector nr_puncte x 2)
    for j = 2:nr_puncte.
        for i = 1:nr_puncte - j + 1.
            P(i,:) = (1-t).P(i,:) + t.P(i,:).
        endfor
    endfor.
    P_max = P(1,:).
endfunction
```

# Aproximare Numerică

* **Spațiu Prehilbertian**: $(F, u)$ → $F$ = spațiu vectorial; $u$ = produs scalar

$u: F \times F \to \mathbb{R}$, $u(f_1, f_2) = \langle f_1, f_2 \rangle$

Ex. $F = \mathbb{R}^n$, $u = \langle x, y \rangle = \sum_{i=1}^{n} x_i y_i$, $x_i, y_i \in \mathbb{R}^n$

$F = \mathbb{C}[a,b]$, $u = \langle f_1, f_2 \rangle = \int_a^b f_1(x) \cdot f_2(x) \cdot w(x)\, dx$, $f_1, f_2 \in \mathbb{R}^n$

Proprietățile produsului scalar:

- $\langle f_1, f_2 \rangle = \langle f_2, f_1 \rangle$
- $\langle f_1 + f_2, f_3 \rangle = \langle f_1, f_3 \rangle + \langle f_2, f_3 \rangle$
- $\langle \alpha f_1, f_2 \rangle = \alpha \langle f_1, f_2 \rangle$
- $\langle f_1, f_2 \rangle \geq 0$.

Fie $G$ - un subspațiu de dimensiune finită al lui $F$.




$f \in F \xrightarrow[\text{în sensul c.m.m.p.}]{\text{cel mai bun aproximant}} g^* \in G$ a.i. $\|f - g^*\|_2 = \min_{g \in G} \{\|f - g\|_2\}$.

Condiție necesară și suficientă pentru ca $g^*$ să fie cel mai bun aproximant în sensul c.m.m.p pentru $f$:

$\langle f - g^*, g \rangle = 0$, $\forall g \in G$.

Fie $\{u_0, u_1, \dots, u_n\}$ bază pentru $G$. $\Rightarrow g^* = \sum_{k=0}^{n} c_k^* u_k$, $c_k^* = ?$

$\sum_{k=0}^{n} c_k^* \langle u_j, u_k \rangle = \langle u_j, f \rangle$, $\forall j = 0:n$. $\Rightarrow$

$$\Rightarrow \begin{bmatrix} \langle u_0, u_0 \rangle & \langle u_0, u_1 \rangle & \dots & \langle u_0, u_n \rangle \\ \vdots & & & \vdots \\ \langle u_n, u_0 \rangle & \langle u_n, u_1 \rangle & \dots & \langle u_n, u_n \rangle \end{bmatrix} \begin{bmatrix} c_0^* \\ \vdots \\ c_n^* \end{bmatrix} = \begin{bmatrix} \langle u_0, f \rangle \\ \vdots \\ \langle u_n, f \rangle \end{bmatrix}.$$

Baza ortogonală: $\langle u_i, u_j \rangle = 0$, $i \neq j \Rightarrow c_k^* = \dfrac{\langle u_k, f \rangle}{\langle u_k, u_k \rangle}$, $k = 0:n$

Baza ortonormată: $\langle u_i, u_j \rangle = 0$, $i \neq j$; $\langle u_i, u_j \rangle = 1$, $i = j$ $\Rightarrow c_k^* = \langle u_k, f \rangle$, $k = 0:n$.

# Aproximare Continuă în sensul CMMP

- **Produsul scalar în cazul continuu:** $\langle f,g\rangle = \int_a^b f(x)\cdot g(x)\cdot w(x)\,dx$. $f,g \in C[a,b]$, $w$ funcție pondere.
- **Baza polinomială:** $\{1, x, x^2, \dots, x^n\}$.

  $g^*(x) = c_n^* x^n + c_{n-1}^* x^{n-1} + \dots + c_1^* x + c_0^*$

  * se mai notează diferit matricial
- **Baza trigonometrică** ($[a,b]=[0,2\pi]$, $w(x)=1$): $\{\frac{1}{\sqrt{2}}, \cos x, \sin x, \dots, \cos nx, \sin nx\}$.

  $g^*(x) = \frac{1}{\sqrt{2}} a_0 + \sum_{p=1}^{n} (a_p \cos px + b_p \sin px)$.

  $a_0 = \langle \frac{1}{\sqrt{2}}, f\rangle = \frac{1}{\pi}\int_0^{2\pi} f(x)\cdot\frac{1}{\sqrt{2}}\,dx$.

  $a_p = \langle \cos px, f\rangle = \frac{1}{\pi}\int_0^{2\pi} f(x)\cdot\cos px\,dx$, $\forall\, p=1:n$.

  $b_p = \langle \sin px, f\rangle = \frac{1}{\pi}\int_0^{2\pi} f(x)\cdot\sin px\,dx$, $\forall\, p=1:n$.

  Obs. $\frac{1}{\pi}\int_0^{2\pi} u_i(x)\,u_j(x)\,dx = \begin{cases} 0, & i \neq j \\ 1, & i = j \end{cases}$.
- **Baza Cebâșev** ($[a,b]=[-1,1]$, $w(x)=\frac{1}{\sqrt{1-x^2}}$): $\{\frac{1}{\sqrt{2}}, T_1(x), \dots, T_n(x)\}$.

  Polinomul Cebâșev de ordin $n$: $T_n(x) = \cos(n \arccos x)$, $n \geq 0$.

  $T_{n+1}(x) + T_{n-1}(x) = 2x\,T_n(x)$, $T_0(x)=1$, $T_1(x)=x$.

  $g^*(x) = \frac{1}{\sqrt{2}} a_0 + \sum_{p=1}^{n} a_p T_p(x)$.

  $a_0 = \langle \frac{1}{\sqrt{2}}, f\rangle = \frac{1}{\pi}\int_{-1}^{1} \frac{f(x)\cdot\frac{1}{\sqrt{2}}}{\sqrt{1-x^2}}\,dx$.

  $a_p = \langle T_p, f\rangle = \frac{2}{\pi}\int_{-1}^{1} \frac{f(x)\cdot T_p(x)}{\sqrt{1-x^2}}\,dx$, $\forall\, p=1:n$.

  Obs. $\int_{-1}^{1} \frac{T_i(x)\cdot T_j(x)}{\sqrt{1-x^2}}\,dx = \begin{cases} 0, & i \neq j \\ \frac{\pi}{2}, & i=j\neq 0 \\ \pi, & i=j=0 \end{cases}$.

---

Exemplu Rezolvat

$f(x)=x$, $x\in[-1;1]$, $w(x)=1$

Aproximați $f(x) \simeq P_2(x)$

Baza polinomială: $\{1, x, x^2\}$

$P_2(x) = c_2^* x^2 + c_1^* x + c_0^*$

$$\begin{bmatrix} \langle 1,1\rangle & \langle 1,x\rangle & \langle 1,x^2\rangle \\ \langle x,1\rangle & \langle x,x\rangle & \langle x,x^2\rangle \\ \langle x^2,1\rangle & \langle x^2,x\rangle & \langle x^2,x^2\rangle \end{bmatrix} \begin{bmatrix} c_0^* \\ c_1^* \\ c_2^* \end{bmatrix} = \begin{bmatrix} \langle 1,x\rangle \\ \langle x,x\rangle \\ \langle x^2,x\rangle \end{bmatrix} \Leftrightarrow$$

$$\Leftrightarrow \begin{bmatrix} 2 & 0 & \frac{2}{3} \\ 0 & \frac{2}{3} & 0 \\ \frac{2}{3} & 0 & \frac{2}{5} \end{bmatrix} \begin{bmatrix} c_0^* \\ c_1^* \\ c_2^* \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{2}{3} \\ 0 \end{bmatrix} \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} 2c_0^* + \frac{2}{3}c_2^* = 0 \\ \frac{2}{3}c_1^* = \frac{2}{3} \\ \frac{2}{3}c_0^* + \frac{2}{5}c_2^* = 0 \end{cases} \Rightarrow \begin{cases} c_0^* = 0 \\ c_1^* = 1 \\ c_2^* = 0 \end{cases}$$

---

Exemplu Rezolvat

$f(x)=x$, $x\in[-\pi;\pi]$, $w(x)=1$

Aproximați $f(x)$ în mod trigonometric ($n=1$)

Baza trigonometrică: $\{\frac{1}{\sqrt{2}}, \cos x, \sin x\}$

$g^*(x) = \frac{a_0}{\sqrt{2}} + \sum_{p=1}^{n} (a_p \cos px + b_p \sin px)$

$a_0 = \langle \frac{1}{\sqrt{2}}, x\rangle = \frac{1}{\pi}\int_{-\pi}^{\pi} x\cdot\frac{1}{\sqrt{2}}\,dx = 0$

$a_1 = \langle \cos x, x\rangle = \frac{1}{\pi}\int_{-\pi}^{\pi} x\cos x\,dx = 0$

$b_1 = \langle \sin x, x\rangle = \frac{1}{\pi}\int_{-\pi}^{\pi} x\sin x\,dx = 2$

# Aproximare Discretă în sensul C.M.M.P.

- Produsul scalar în cazul discret: $\langle f, g \rangle = \sum_{i=0}^{n} f(x_i) g(x_i) \cdot w(x_i)$.

$$\|f - g^*\|_2 = \min_{g \in G} \{\|f - g\|_2\} = \min_{g \in G} \{\sqrt{\langle f-g, f-g \rangle}\} = \min_{g \in G} \left\{\sqrt{\sum_{i=1}^{n} [f(x_i) - g(x_i)]^2 \cdot w(x_i)}\right\}$$

- Baza polinomială: $\{1, x, \dots, x^n\}$

$g^*(x) = c_n^* x^n + c_{n-1}^* x^{n-1} + \dots + c_1^* x + c_0^*$

$\Rightarrow$ nr. coeficienți din matrice

- Baza trigonometrică: $\left\{\frac{1}{\sqrt{2}}, \cos x, \sin x, \dots, \cos nx, \sin nx\right\}$

$$g^*(x) = \frac{1}{\sqrt{2}} a_0 + \sum_{p=1}^{n} (a_p \cos px + b_p \sin px)$$

$$a_0 = \left\langle \frac{1}{\sqrt{2}}, f \right\rangle = \sum_{k=0}^{2n-1} f(x_k) \cdot \frac{1}{\sqrt{2}}$$

$$a_p = \langle \cos px, f \rangle = \sum_{k=0}^{2n-1} f(x_k) \cdot \cos p x_k$$

$$b_p = \langle \sin px, f \rangle = \sum_{k=0}^{2n-1} f(x_k) \cdot \sin p x_k$$

$$x_k = \frac{k\pi}{n}; \; k = 0 : 2n-1$$

$$\text{Obs. } \sum_{k=0}^{2n-1} u_i(x_k) \cdot u_j(x_k) = \begin{cases} 0, & i \neq j \\ 1, & i = j \end{cases}$$

- Baza cebâșev: $(w(x) = 1)$. $\left\{\frac{1}{\sqrt{2}}; T_1(x); \dots; T_n(x)\right\}$.

$$g^*(x) = \frac{1}{\sqrt{2}} a_0 + \sum_{p=1}^{n} a_p \cdot T_p(x).$$

$$a_0 = \left\langle \frac{1}{\sqrt{2}}, f \right\rangle = \frac{2}{n+1} \cdot \sum_{k=0}^{n} f(x_k) \cdot \frac{1}{\sqrt{2}}$$

$$a_p = \langle T_p, f \rangle = \frac{2}{n+1} \cdot \sum_{k=0}^{n} f(x_k) \cdot T_p(x_k)$$

$$x_k = \cos \frac{(2k+1)\pi}{2n+2}, \; k = 0 : n.$$

$$\text{Obs. } \sum_{k=0}^{n} u_i(x_k) \cdot u_j(x_k) = \begin{cases} 0, & i \neq j \\ \frac{n+1}{2}, & i = j \end{cases}$$

## Exemplu Rezolvat (Examen 2019, Subiectul 7)

| $x$ | $-1$ | $-0{,}5$ | $0$ | $0{,}5$ | $1$ |
|---|---|---|---|---|---|
| $f(x)$ | $1$ | $0{,}5$ | $0$ | $0{,}5$ | $1$ |

$$P_2(x) = C_2^* \cdot x^2 + C_1^* \cdot x + C_0^*$$

$$\begin{bmatrix} \langle 1,1 \rangle & \langle 1,x \rangle & \langle 1,x^2 \rangle \\ \langle x,1 \rangle & \langle x,x \rangle & \langle x,x^2 \rangle \\ \langle x^2,1 \rangle & \langle x^2,x \rangle & \langle x^2,x^2 \rangle \end{bmatrix} \begin{bmatrix} C_0^* \\ C_1^* \\ C_2^* \end{bmatrix} = \begin{bmatrix} \langle 1,f \rangle \\ \langle x,f \rangle \\ \langle x^2,f \rangle \end{bmatrix} \Leftrightarrow$$

$$\Leftrightarrow \begin{bmatrix} \sum_0^4 1 = 5 & \sum_0^4 x = 0 & \sum_0^4 x^2 = 2{,}5 \\ \sum_0^4 x = 0 & \sum_0^4 x^2 = 2{,}5 & \sum_0^4 x^3 = 0 \\ \sum_0^4 x^2 = 2{,}5 & \sum_0^4 x^3 = 0 & \sum_0^4 x^4 = 2{,}125 \end{bmatrix} \begin{bmatrix} C_0^* \\ C_1^* \\ C_2^* \end{bmatrix} = \begin{bmatrix} \sum_0^4 f(x) = 3 \\ \sum_0^4 x \cdot f(x) = 0 \\ \sum_0^4 x^2 \cdot f(x) = 2{,}25 \end{bmatrix} \Rightarrow$$

$$\Rightarrow \begin{cases} 5C_0^* + 2{,}5C_2^* = 3 \\ 2{,}5C_1^* = 0 \\ 2{,}5C_0^* + 2{,}125C_2^* = 2{,}25 \end{cases} \Rightarrow \begin{cases} C_0 = \frac{6}{35} \\ C_1^* = 0 \\ C_2 = \frac{1{,}5}{1{,}75} = \frac{6}{7} \end{cases}$$

# Derivare Numerică

$f: \mathbb{R} \to \mathbb{R}$. $f(x) \simeq P_n(x)$ - polinom de interpolare Lagrange.

- $P_n(x) = \sum_{k=0}^{n} f(x_k) \cdot L_k(x)$.

- $\boxed{n=1} \Rightarrow \{x_0, x_1\}$ - suport de interpolare; $f(x) \simeq P_1(x)$.

  $f'(x_0) \simeq \frac{1}{\ell}[f(x_0+\ell) - f(x_0)]$, unde $\ell$ - lungimea intervalului $[x_0, x_1]$.

- $\boxed{n=2} \Rightarrow \{x_0, x_1, x_2\}$ - suport de interpolare; $f(x) \simeq P_2(x)$.

  $f'(x_0) \simeq \frac{1}{\ell}\left[-\frac{3}{2}f(x) + 2f(x_0+\ell) - \frac{1}{2}f(x_0+2\ell)\right]$, unde $\ell$ - lungimea intervalului $[x_0, x_1]$ $[x_1, x_2]$

  $f'(x_0) \simeq \frac{1}{2\ell}[-f(x_0-\ell) + f(x_0+\ell)]$, unde $\ell$ - lungimea intervalului $[x_0, x_1]$ $[x_1, x_2]$.

  $f''(x_0) \simeq \frac{1}{\ell^2}[f(x_0-\ell) - 2f(x_0) + f(x_0+\ell)]$, unde $\ell$ - lungimea intervalului $[x_0, x_1]$ $[x_1, x_2]$.

# INTEGRARE NUMERICĂ - OCTAVE

$f:[a,b]\to\mathbb{R}$ - continuă, $f(x)\approx P_n(x)$ - polinom de interpolare Lagrange.

- $\int_a^b f(x)\,dx \approx \sum_{k=0}^{n} a_k f(x_k)$, unde $x_k$ - puncte echidistante în $[a,b]$.

$\left(x_k = a + kl,\ k=\overline{0,n},\ \text{unde } l=\frac{b-a}{n}\right.$ - nr. de intervale; $\left.l\right.$ - lungimea fiecărui interval.$\left.\right)$

- $a_k = \int_a^b L_k(x)\,dx$.

## ✱ METODA SIMPLĂ A TRAPEZELOR.

$\boxed{n=1}$ $\{x_0, x_1\}$ - suport de interpolare. $(x_0=a;\ x_1=b \mid l=b-a)$

$\int_a^b f(x)\,dx = \frac{l}{2}\left[f(\underbrace{a}_{x_0}) + f(\underbrace{b}_{x_1})\right],\ l=b-a.$

## ✱ FORMULA SIMPLĂ SIMPSON

$\boxed{n=2}$ $\{x_0, x_1, x_2\}$ - suport de interpolare. $\left(x_0=a;\ x_1=\frac{a+b}{2};\ x_2=b \mid l=\frac{b-a}{2}\right)$.

METODA 1: $\int_a^b f(x)\,dx = \int_a^b P_2(x)\,dx$

METODA 2: $\int_a^b f(x)\,dx \approx \frac{l}{3}\left[f(\underbrace{a}_{x_0}) + 4f(\underbrace{\tfrac{a+b}{2}}_{x_1}) + f(\underbrace{b}_{x_2})\right],\ l=\frac{b-a}{2}$.

## ✱ FORMULA COMPUSĂ A TRAPEZELOR. ($n$ - impar)

$\int_a^b f(x)\,dx = \int_{x_0}^{x_1} f(x)\,dx + \ldots + \int_{x_{n-1}}^{x_n} f(x)\,dx \approx \frac{l}{2}\left[f(x_0) + 2\sum_{i=1}^{n-1} f(x_i) + f(x_n)\right],\ l=\frac{b-a}{n}$.

## ✱ FORMULA COMPUSĂ SIMPSON. ($n$ - par).

$\int_a^b f(x)\,dx = \int_{x_0}^{x_2} f(x)\,dx + \ldots + \int_{x_{n-2}}^{x_n} f(x)\,dx \approx \frac{l}{3}\left[f(x_0) + 4\sum_{i=1}^{\frac{n}{2}} f(x_{2i-1}) + 2\sum_{i=1}^{\frac{n}{2}-1} f(x_{2i}) + f(x_n)\right],\ l=\frac{b-a}{n}$

# INTEGRALE DUBLE

OCTAVE

$f:[a,b]\times[c,d]\to\mathbb{R}$; $I=\int_a^b\int_c^d f(x,y)\,dy\,dx$

$I_1=\int_c^d f(x,y)\,dx \overset{\text{Simpson simplu}}{\simeq} \frac{d-c}{2\cdot 3}\left[f(x,c)+4f\left(x,\frac{c+d}{2}\right)+f(x,d)\right]$

$I_2=\frac{d-c}{6}\int_a^b f(x,c)+4f\left(x,\frac{c+d}{2}\right)+f(x,d)\,dx \overset{\text{Simpson simplu}}{\simeq}$

$\simeq \frac{d-c}{6}\cdot\frac{b-a}{2\cdot 3}\left[f(a,c)+4f\left(\frac{a+b}{2},c\right)+f(b,c)+f\left(a,\frac{c+d}{2}\right)+4f\left(\frac{a+b}{2},\frac{c+d}{2}\right)+f\left(b,\frac{c+d}{2}\right)+f(a,d)+4f\left(\frac{a+b}{2},d\right)+f(b,d)\right]$

# INTEGRARE NUMERICĂ A CUADRATURII GAUSSIENE

$$\int_a^b f(x)\,w(x)\,dx \simeq \sum_{k=0}^{n} \sigma_k f(x_k).$$

| FAMILIE DE POLINOAME ORTOGONALE | $a$ | $b$ | $w(x)$ | RELAȚIE DE RECURENȚĂ |
|---|---|---|---|---|
| CEBÂȘEV DE ORDINUL 1 | $-1$ | $1$ | $\frac{1}{\sqrt{1-x^2}}$ | $T_{n+1}-2xT_n+T_{n-1}=0$ $[T_0=1,\ T_1=x]$ |
| CEBÂȘEV DE ORDINUL 2 | $-1$ | $1$ | $\sqrt{1-x^2}$ | $T_{n+1}-2xT_n+T_{n-1}=0$ $[T_0=1,\ T_1=2x]$ |
| LEGENDRE | $-1$ | $1$ | $1$ | $(n+1)L_{n+1}-(2n+1)xL_n+nL_{n-1}=0$ $[L_0=1,\ L_1=x]$ |
| LAGUERRE | $0$ | $\infty$ | $e^{-x}$ | $G_{n+1}-(2n+1-x)G_n+n^2G_{n-1}=0$ $[G_0=1,\ G_1=1-x]$ |
| HERMITE | $-\infty$ | $\infty$ | $e^{-x^2}$ | $H_{n+1}-2xH_n+2nH_{n-1}=0$ $[H_0=1,\ H_1=2x]$ |

# Metode Numerice cu Pași Separați

OCTAVE

$I=[a,b]$; $y: I \to \mathbb{R}$, $f: I \times \mathbb{R} \to \mathbb{R}$.

**Problema diferențială de ordin 1**

$y'(t) = f(t, y(t))$ ; $y(t) = ?$

**Problema Cauchy de ordin 1.**

$y'(t) = f(t, y(t))$, $y(t_0) = \lambda$ ; $y(t) = ?$

* se împarte echidistant: $t_k = a + k \cdot l$, $k = 0:n$, $l = \frac{b-a}{n}$ ; $\Rightarrow y(t_i) = ?$

lungimea intervalului — numărul de intervale.

- **Metoda Euler (Runge-Kutta de ordin 1)**

$$y(t) = y(t_i) + (t-t_i)y'(t_i) + \frac{(t-t_i)^2}{2} y''(t_i) + \dots$$

$t = t_{i+1} \Rightarrow y(t_{i+1}) = y(t_i) + (t_{i+1}-t_i)y'(t_i) + \dots$ (PROBLEMA CAUCHY.)

Fie $y_i \leftarrow y(t_i) \Rightarrow y_{i+1} = y_i + l \cdot f(t_i, y_i)$; $i = 0:n$; $y_0 = \lambda$, $l = \frac{b-a}{n}$.

- **Metoda Runge-Kutta de ordin 2.**

$y_{i+1} = y_i + \frac{K_1 + K_2}{2}$; $i = 0:n$; $y_0 = \lambda$

$K_1 = l \cdot f(t_i, y_i)$, $l = \frac{b-a}{n}$.

$K_2 = l \cdot f(t_i + l, y_i + K_1)$; $l = \frac{b-a}{n}$

- **Metoda Runge-Kutta de ordin 4.**

$$y_{i+1} = y_i + \frac{K_1 + 2K_2 + 2K_3 + K_4}{6}$$

$K_1 = l \cdot f(t_i, y_i)$, $l = \frac{b-a}{n}$

$K_2 = l \cdot f(t_i + \frac{l}{2}, y_i + \frac{K_1}{2})$, $l = \frac{b-a}{n}$

$K_3 = l \cdot f(t_i + \frac{l}{2}, y_i + \frac{K_2}{2})$, $l = \frac{b-a}{n}$

$K_4 = l \cdot f(t_i + l, y_i + K_3)$, $l = \frac{b-a}{n}$.

# METODE NUMERICE CU PAȘI LEGAȚI.

OCTAVE

$y'(t) = f(t, y(t))$, $y(t_0) = \lambda$; $y(t) = ?$

Fie $y_i = y(t_i)$.

$f_i = f(t_i, y_i) = f(t_i, y(t_i))$.

- **METODA ADAMS-BASHFORTH (METODA EXPLICITĂ).**

$y_{i+1} = y_i + l \sum_{j=0}^{r-1} \beta_j \cdot f_{i-j}$, $y_0 = \lambda$.

$r$ - ordinul metodei A-B.

DETERMINARE $\beta_j$, $j = 0:r-1$.

- se alg punctele $t_{i-r+1}, \dots, t_i, t_{i+1}$ echidistante cu $l = 1$, și $t_{i-r+1} = 0$.
- se pun condițiile ca $y_{i+1} = y_i + l \cdot \sum_{j=0}^{r-1} \beta_j f_{i-j}$ să fie exacte.

dacă $y(t) = \{1, t, \dots, t^r\}$.

- **METODA ADAMS-MOULTON (METODA IMPLICITĂ).**

$y_{i+1} = y_i + l \cdot \sum_{j=-1}^{r-1} \beta_j \cdot f_{i-j}$.

$r$ - ordinul metodei A-M.

DETERMINARE. $\beta_j$, $j = -1:r-1$.

- se alg punctele $t_{i-r+1}, \dots, t_{i+1}$ echidistante cu $l = 1$ și $t_{i-r+1} = 0$.
- se pun condițiile ca $y_{i+1} = y_i + l \cdot \sum_{j=-1}^{r-1} \beta_j \cdot f_{i-j}$ să fie exacte

dacă $y(t) = \{1, t, \dots, t^{r+1}\}$.

- **METODA PREDICTOR - CORECTOR. (A-B-M).**

- A-B: se folosește pentru predicția unei valori $y_{i+1}$.
- A-M: se folosește pentru corectarea acelei valori.